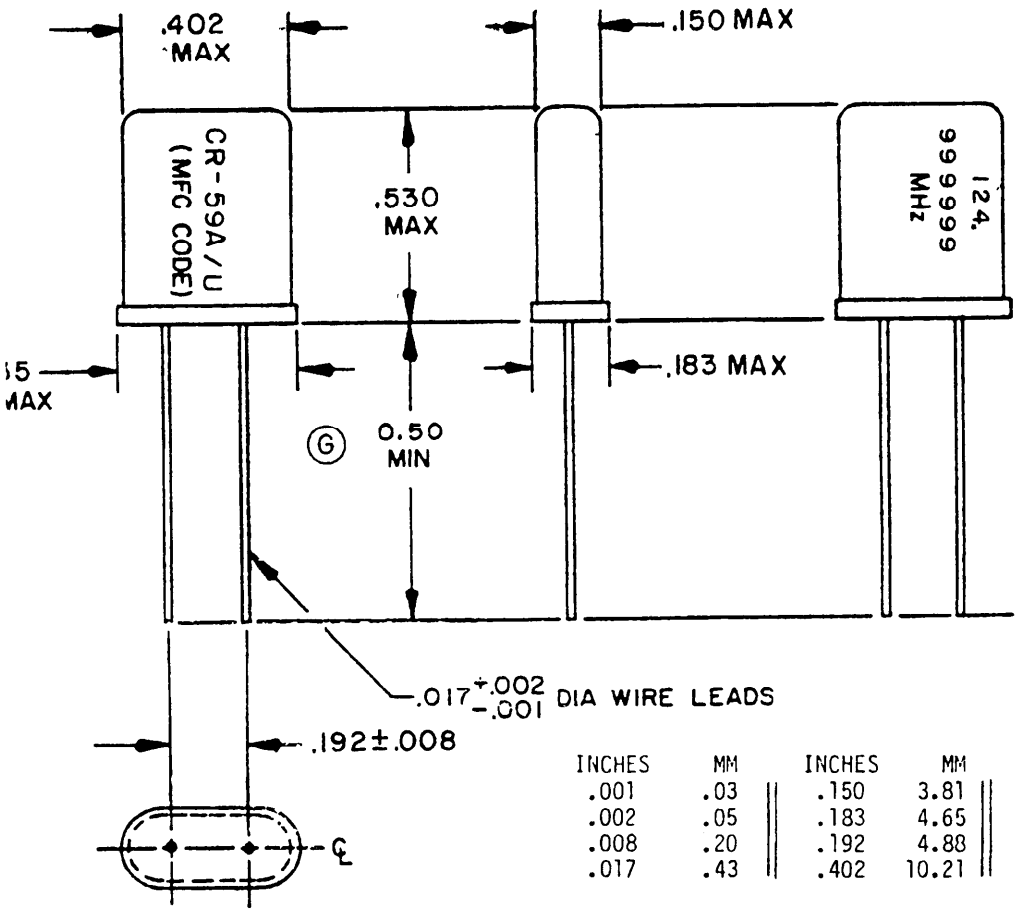
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-59A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth overtone; controlled; series.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.50	12.7
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Dimensions and configuration.

Ⓖ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
(G) Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure
Frequency range	50 to 125 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerances:	
Operating temperature range	±20 PPM
Room temperature	±80 PPM
Frequency stability	±5 PPM
Equivalent resistance:	Ohms, maximum
50 to 100 MHz	50
100 to 125 MHz	60
Mode of oscillation	Fifth mechanical overtone
Reference temperature	85°C ±1°C
Temperature ranges:	
Operating	80°C to 90°C, inclusive
Operable	-55°C to +80°C, inclusive
Government reference standard test set	TS-683/TSM or AN/TSM-15 (above 100 MHz)
Calibration values:	
Resistance	60 ohms
Resistor voltage drop	0.24 volt
Drive level, rated	1.0 ±0.2 mW
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 PPM

1/ Method 201 of MIL-STD-202.

NOTE: NATO preferred type NXT20/A.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

Users activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-3)

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